Reyaa Divakar

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RESEARCH INTERESTS & PROFILE

Undergraduate Mechanical Engineering student interested in Autonomous Robotics, Mechatronics, Dynamic Controls, and Embedded Hardware. Eager to explore physical robot design, sensor fusion, and control algorithms, leveraging active experience in lab research, hardware fabrication, microcontrollers, and CAD to advance practical robotic systems.

EDUCATION

Colorado School of Mines Aug 2026 – May 2029
B.S. in Mechanical Engineering Golden, CO
Concentration: Automation and Controls
Relevant Coursework: MEGN200 (Programming & Hardware Integration), MEGN201 (Design & Fabrication)

Arapahoe Community College Aug 2022 – May 2026
Concurrent Coursework – Focus: Business & Robotics Engineering

Rock Canyon High School Aug 2022 – May 2026
High School Diploma – Focus Pathway: Mechatronics, Robotics, and Automation Engineering

TECHNICAL SKILLS & LANGUAGES

- **Programming:** MATLAB, Python, Java, HTML/CSS, JavaScript, C#
- **Hardware & Tools:** Arduino, Raspberry Pi, Sensors, Multimeters, Soldering, Circuit Assembly
- **CAD & Software:** SolidWorks, OnShape, OpenCV, MATLAB, Unity, Vercel, Git, Figma
- **Languages:** English (Bilingual), Tamil (Bilingual), Chinese (Limited Working), Spanish (Elementary)

ACADEMIC RESEARCH EXPERIENCE

MIRRORLabs (Mines Interactive Robotics Research) Aug 2026 – Present
Undergraduate Research Assistant Colorado School of Mines, Golden, CO

- Conduct physical robotics fabrication, structural assembly, and hardware testing for trauma-informed HRI research evaluating human psychological responses to robotic movement.
- Collaborate on physical fabrication and mechatronics assembly for a Fungi Learning Robot platform targeting submission to the HRI Student Design Competition as co-author.
- Execute hardware troubleshooting routines, log sensor calibration data, and assist in drafting technical research documentation for empirical lab trials.

PROFESSIONAL & INDUSTRY EXPERIENCE

Burns & McDonnell Jun 2025 – Jul 2025
Summer Shadow Program Intern

- Participated in technical workshops, hardware fabrication sessions, and electrical soldering modules.
- Shadowed senior engineers across multidisciplinary plant design, controls infrastructure, and field engineering workflows.

Colorado Space Business Roundtable (CSBR/ASA) May 2024 – Jul 2024
Aerospace Intern

- Engaged in aerospace industry immersion, learning directly from executive leaders and systems engineers about commercial space projects, business operations, and engineering disciplines.

FlexContent (formerly Investor Radar) Sep 2023 – Feb 2024
Client Acquisition & Marketing Intern

- Executed B2B client acquisition research, drafted SEO-optimized technical blog posts, and tracked campaign analytics performance to refine conversion rates.

Jetson Jun 2023 – Aug 2023

Jetson Project Intern (Product & Tech Focus Group)

- Selected for a highly competitive 10-week summer startup accelerator cohort (∼7% acceptance rate out of 2,000+ national applicants). Executed Figma prototype to pitch my business product/app.
- Conducted QA testing on mobile beta builds, identifying critical user-flow bugs to achieve 0 reported application bugs in live releases.
- Collaborated on competitive analysis, target MVP building, and feature strategies that increased daily active engagement by +5% and day-one retention by +9% overall.

TECHNICAL PROJECTS

Mines Cornerstone: The Kraken Sound Sensor System Aug 2026 – Present

Sensors, Microcontrollers, CAD

- Co-engineering an acoustic search-and-rescue sensing platform to detect human voice frequencies under rubble, being prepared for an upcoming Mines design competition.
- Integrating acoustic Arduino sensors and filtering pipelines to isolate vocal frequencies from environmental background noise during emergency operations and unique robotic design.

Lucky Charm AI Vision Assistant Jun 2026 – Present

Python, OpenCV, Raspberry Pi, Computer Vision

- Engineering an ongoing personal project focused on a computer vision assistant using Python and OpenCV running on a Raspberry Pi hardware stack.
- Developing real-time frame extraction loops to process optical camera data, highlight target visual attributes, and provide dynamic user feedback.

TSA UAV Drone with Mechanical Claw Dec 2024 – Feb 2025

Drone Fabrication, Mechanical Design

- Designed and built a custom uncrewed aerial vehicle (UAV) equipped with an integrated mechanical claw payload for obstacle recovery tasks.
- Earned 3rd Place (Districts Bronze) in the TSA Drone Challenge UAV competition.

Code Connect Non-Profit Platform Jan 2024 – Aug 2025

JavaScript, HTML/CSS, Git, Vercel Deployment

- Designed, built, and deployed the official web platform for Code Connect Edu, hosted via Vercel with version control managed through GitHub.

Internal Reflection (TSA Video Game) Aug 2024 – Jul 2025

Unity, Digital Animation, Team Collaboration

- Collaborated in a multidisciplinary team to design and program a two-player cooperative video game focused on communication and puzzle mechanics in Unity.
- Awarded 1st Place State and 3rd Place Nationally at the Technology Student Association (TSA) National Conference (2025).

Digital Media & YouTube Channels Aug 2023 – Present

Video Editing, Creative Content/Storytelling

- Manage and produce content across personal YouTube channels covering humor, creative digital media, gaming, art, and tech.

LEADERSHIP & COMMUNITY IMPACT

Nexus Language Jan 2024 – Present

Paid On-Site Tamil Interpreter

- Employed as a paid professional on-site Tamil interpreter for educational institutions and Child Find developmental screening programs across Colorado.
- Deliver real-time translation during IEP meetings, parent conferences, and pediatric evaluations.

Code Connect Aug 2023 – Aug 2026
Founder

- Co-founded and lead an educational non-profit organization dedicated to providing accessible computer science training to middle school students.
- Authored curriculum modules and taught 70+ instructional hours covering Python, HTML/CSS, web design, and computational problem-solving to 80+ students.

Tamil Association of Colorado Jan 2024 – May 2026
Youth Committee Designer & Lead

- Initiated and grew the organization's official Instagram social media presence using SEO strategies to boost community engagement.
- Designed print and digital promotional materials using Canva to establish consistent visual identity for cultural events.

HONORS & AWARDS

- STEM Highest Honors Student – Rock Canyon High School (RCHS), (May 2026)
- Outstanding Technology & Engineering Student of the Year – RCHS Departmental Award, (May 2026)
- Science Olympiad State 3rd Place – Engineering CAD Modeling, (March 2026)
- Science Olympiad State 2nd Place – Materials Science, (March 2026)
- Science Olympiad District 4th Place – Engineering CAD Modeling, (Feb 2026)
- TSA TEAMS State 2nd Place – Engineering Exam & Practical STEM Challenge, (Jan 2026)
- TSA Districts Conference 3rd Place – Drone Challenge (UAV), (Dec 2025)
- TSA National Conference 3rd Place – Video Game Design, (July 2025)
- TSA State Conference 1st Place – Video Game Design, (Feb 2025)
- TSA District Conference 1st Place – Digital Video Production, (Dec 2023)
- TSA District Conference 2nd Place – Digital Video Production, (Dec 2022)

CERTIFICATIONS & TRAINING

- SolidWorks CSWA Certification (Expected Dec 2026)
- Export Compliance (EC) Training – Academic & Tech Transfer Regulations
- Research Security Training (Combined) – Federal Research Integrity & Protocol Compliance
- Conflicts of Interest (COI) Certification – Academic Research Standards
- 525F Multimeter Electrical Hardware Certification – Electrical Diagnostics & Safety
- Initial & Grade Three Electric Keyboard Certifications – Trinity College London (May 2022)